PROBLEM 2

Revolutionising Dominican Republic’s energy system

High School - Scholars
Team# 7213 - MegaMinds

Gayatri Papagari, Hima Keerthi Raparla, Maya Yadlapalli, Nirvaan Duggirala,

THE FACTS

11M People

Affected in a nationwide blackout in November 2025 after a transmission failure cascaded across the grid.

Over 80%

Of the country's electricity, according to LowCarbonPower, is generated with fossil fuels and is largely dependent on imported oil, leading to energy dependence.

$1.2B Lost

In distribution-and transmission during 2017-2021, which is 1.4% of the country's GDP (according to the World Bank)

THE STAKEHOLDERS



THE GOVERNMENT OF THE DOMINICAN REPUBLIC



CITIZENS OF THE DOMINICAN REPUBLIC



PRIVATE ENERGY COMPANIES THAT ACCOUNTS FOR 86% OF ENERGY PRESENTATION

THE

PROBLEM

1.4M	10-25%	11.6%

Weather Impacts

Power lines, substations, and generation sites suffer severe hurricane impacts. In 2022, Hurricane Fiona affected **1.4 million** people and left **350,000** without electricity, causing prolonged nationwide outages and infrastructure failures.

Overloaded, & Aging Grid

Much of the grid is outdated and poorly maintained. A World Bank report shows feeders experience **10–25%** voltage drops far above the **5%** standard, indicating overload, instability, and growing demand pressures.

Energy Loss in Distribution

Long-distance transmission overheats power lines and reduces efficiency. CEIC/World Bank data show distribution losses historically around **11.6%**, worsening substation stress and increasing national energy waste and system costs.

GOAL 1
DECENTRALISATION

Vision Statement: Communities throughout the Dominican Republic will produce and oversee their own dependable, reasonably priced energy by 2040 thanks to the widespread use of microgrids, prosumer involvement, and locally managed clean power systems.

Action Plan Overview: Implementation of microgrids in/near human settlements run on renewable energy that can be isolated from central power sources

1. **2026-2028:** Planning of microgrid set-up, energy flow and balance and the associated energy production source for microgrids in chosen settlements.
2. **2029-2032:** Building of microgrids begins in beta settlements with simultaneous testing.
3. **2033-2038:** Microgrid technology is expanded to other settlements and adapted to their needs.
4. **2038-2040:** Nation-wide implementation of microgrids is achieved to the extent that all large human settlement (more than 100,000 people) are decentralised from the major energy plants.

DECENTRALISATION
DETAILED ACTION PLAN

1

Plan settlement-level microgrid designs

Develop detailed microgrid plans for selected communities(especially rural), mapping energy flows, storage needs, and renewable sources to ensure stable, localized power independent of national grid disruptions .

2

Construct and test microgrids in pilot settlements

Begin building microgrids in beta settlements, integrating solar, wind, or hydro as appropriate, while testing performance, reliability, and community energy demand in real time. Pilot microgrids worldwide show **5–10%** improved reliability and major reductions in diesel fuel dependence during emergencies.

3

Expand microgrid systems to additional communities

Extend microgrid deployment to more settlements between 2033–2038, tailoring each system to local conditions to improve energy access, stability, and climate resilience. In rural / off-grid communities, microgrids using solar + wind + storage have been shown to cut diesel use by up to **70 %**, reducing carbon emissions by **150 tons CO_2/year** in example models.

4

Achieve nationwide microgrid decentralisation

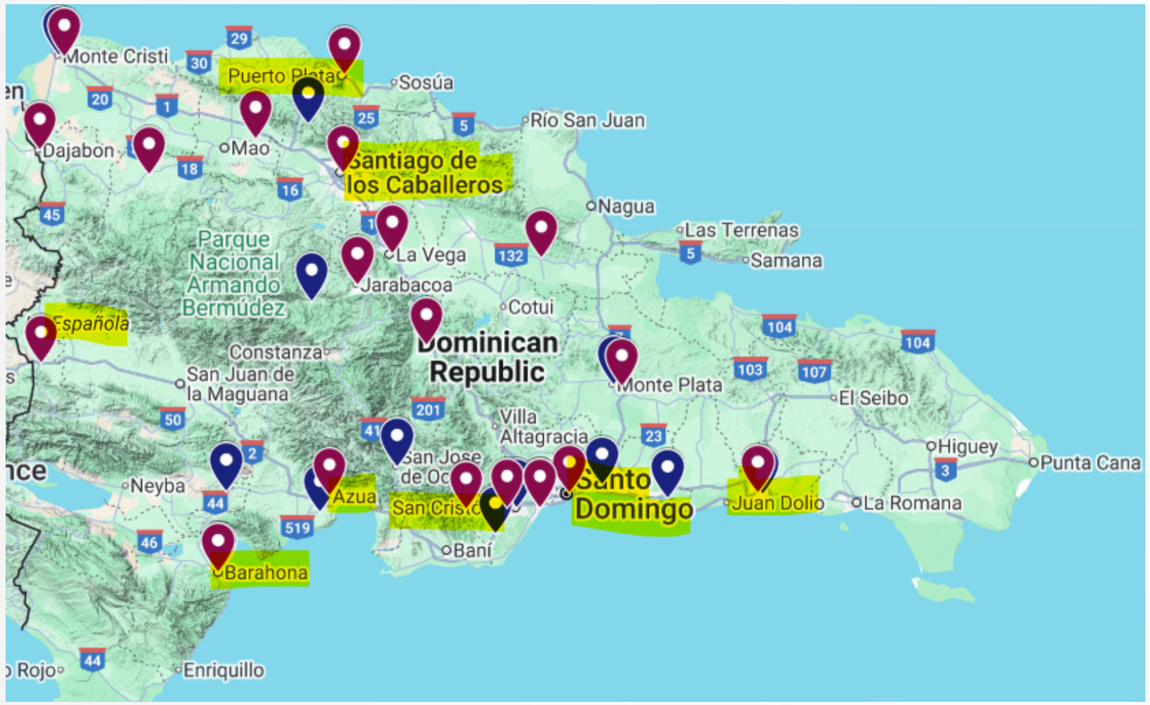
By 2038–2040, establish microgrid networks across all major settlements (100,000+ residents), reducing dependence on centralized power plants . Such expansions have consistently increased community resilience, with renewable-based systems cutting emissions by **over 100 tons of CO_2 annually.**

IMPACT AND COST

Cost Estimates

At this stage, it is impossible to give close cost estimates as that varies wildly from region to region and the way the work proceeds. However, there are some factors that can impact cost:

1. When the microgrid is far from existing power plants, new transmission, extensive site prep and more technological complexity is required
2. When there is a higher disaster risk, it requires extra hardware, segmentation, back-up storage, which costs more



A map of existing energy plants **(blue)** and potential new energy plants **(red)**. Highligted locations have higher disaster risk (mostly coastal areas)

Human Resources



Community Organizations



Industry Partners



NGOs



Technical Experts

Resources needed

Physical Resources+ Price Range Estimates:

1. Solar PV panels, wind turbines, micro-hydro (site dependent)
   a. Solar panel - **US$ 1,300–2,700 / kW**
   b. Wind Turbines(Onshore) - **US$ 900–1,750 / kW**
   c. Micro-hydro - **US$ 1,000–5,000 / kW installed**
2. Advanced battery storage systems and switchgear - **US$ 200–400 / kWh** (or about US$ 800–1,200 / kW for a 4-hour system (according to HighJoule)
3. Reinforced structures (poles, foundations) - **US$ 200–500 / kW**
4. Smart control and communications systems for grid flexibility/islanding - **US$ 50–200 / kW** (price can vary alot)

Even though installing microgrids can initially be expensive to set up, studies show they are highly cost effective over the long term, with **19–35%** lower life-cycle costs and up to **1.16× ROI over 20 years**.

2026–2029: Install new infrastructure and reinforce existing infrastructure. Simultaneously, implement new technology like metereological monitoring.

1

2030–2035: Expand automated islanding systems and ensure battery storage extends to critical locations (hospitals, etc.)

2

2036–2040: Ensure grids are weather preemptive and adaptive by integrating pre-storm isolation softwares.

3

GOAL 2
RESILIENCE

Vision Statement: By 2040, the Dominican Republic will operate a climate-resilient power grid that can withstand severe weather, protect vital services, and reduce the number of hours of national outages by more than half.

Action Plan Overview:

1. Improved technology and infrastructure which includes corrosion-resistant equipment
2. Implementatiion of pre-storm automated islanding technology into existing energy infrastructure
3. Improve coordination and intelligent forecasting using metereological monitoring to initiate early alerts and allow grids to adapt
4. Backup power through energy storage

RESILIANCE
DETAILED ACTION PLAN

1

Improved technology and infrastructure
Corrosion-resistant equipment and better maintenance of weak grid sections reduce storm failures, restoration time, and provide Dominican communities with safer, more reliable electricity during hurricanes. Hardened infrastructure cuts hurricane outages by **40–60%.**

2

Improve coordination and intelligent forecasting
To lessen storm-related disruptions and safeguard Dominican communities, increase meteorological monitoring and coordinate civil protection agencies to initiate early alerts and focused grid actions. Early-warning improve response times by **30–50%.**

3

Pre-storm automated islanding
In order to prevent cascading failures, safeguard weak feeders, and assist Dominican communities in maintaining vital power services during severe weather. Automated islanding isolates susceptible grid segments prior to natural disasters.

4

Backup power through energy storage
In order to maintain critical services and improve grid resilience for Dominican communities during extended outages, batteries and other energy storage devices store excess power for later use even during critical times.

2026–2027: Identify locations for further expansion for wind energy and hydroenergy. Assess national biomass potential.

1

2027–2033: Begin expansion of solar-storage projects and start constructing wind farms and modernizing hydroplants.

2

2033–2037: Plan for the integration of renewable energy into existing microgrids

3

GOAL 3
SUSTAINABILITY

Vision Statement: By 2040, the Dominican Republic will achieve a clean, low-carbon energy system powered by at least 50% renewables, significantly lowering fuel imports, enhancing energy independence, and lowering carbon emissions at the national level.

Action Plan Overview: Improve individual renewable energy production

1. Increase utility-scale solar using battery storage
2. Create specific wind farm projects along the coast
3. Update and renovate small hydroplants
4. Utilise biomass potential

SUSTAINABILITY
DETAILED ACTION PLAN

1

Expanding solar power and using battery storage.
Increase clean generation, decrease fossil fuel imports, and improve grid reliability during extreme weather events by expanding large solar farms with storage to harness the Dominican Republic's abundant solar irradiance.

2

Create specific wind projects along the coast
Coastal wind farms should be strategically placed to safely capture strong regional winds, , lessen susceptibility to fuel interruptions, and increase the nation's overall energy resilience.

3

Update and renovate small hydro plants
Modernise outdated small hydro facilities to boost resilience against storm-related energy disruptions, stabilise rural grids, boost efficiency, and supply steady renewable power during periods of high demand.

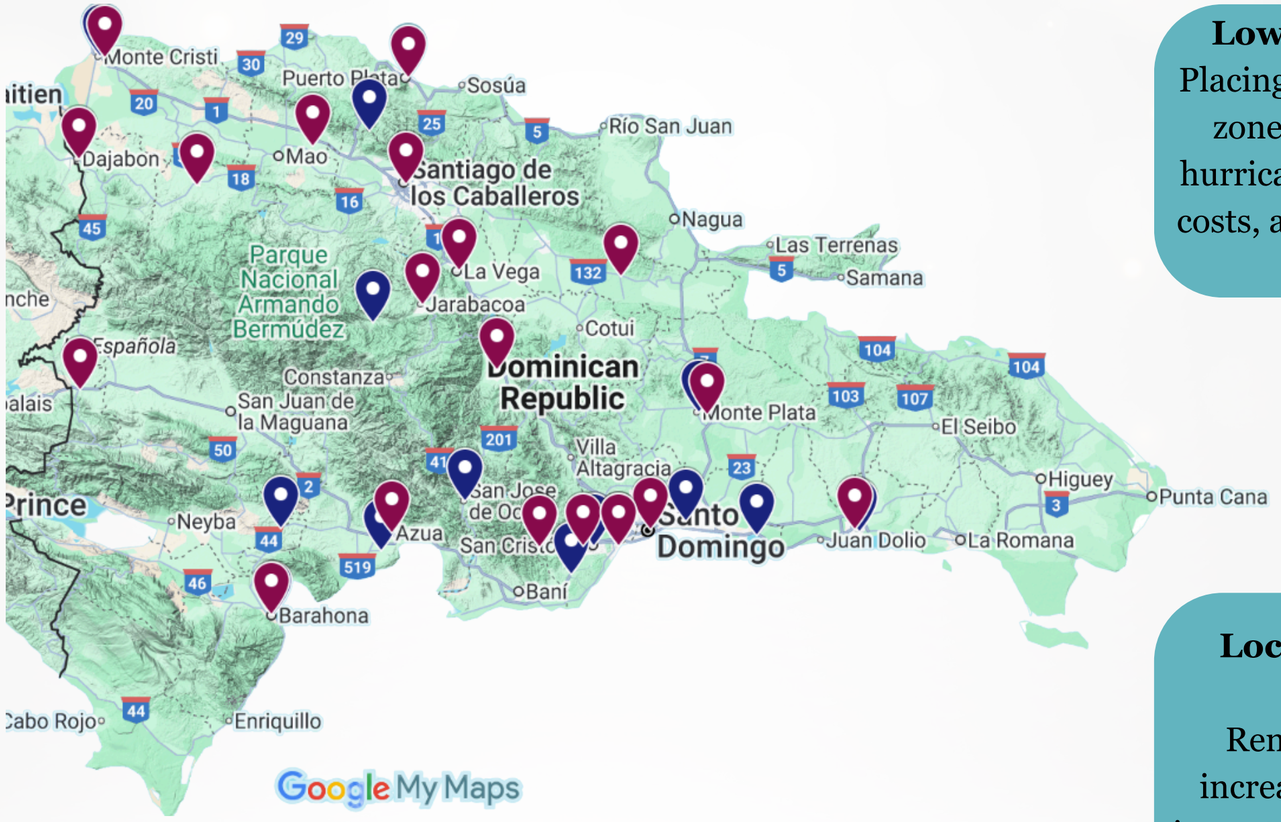
4

Tap into the countries Biomass potential
Create biomass plants with sugarcane residues to help rural economies, diversify power sources, and increase resilience to severe weather and extended outages.

Map Of Potential Renewable Energy Sites

Resource-rich renewable sites
Choosing areas like Montecristi and Valverde with strong sun or wind ensures consistent generation of up to **50 MW** and long-term sustainable growth.

Expanding rural renewable generation
Developing renewable installations in remote regions delivers affordable, reliable electricity to underserved areas; for example, a **40 MW** solar plant can power over **16,000** rural households.



A map of existing energy plants **(blue)** and potential new energy plants **(red)**

Low-risk, resilient locations
Placing projects in lower disaster-risk zones such as Jarabacoa reduces hurricane damage, cuts maintenance costs, and provides reliable power for up to **100,000** people.

Local economic and energy benefits
Renewable projects create jobs, increase local revenue, reduce fuel imports, and provide cleaner, cheaper energy, strengthening long-term community resilience and independence.

IMPACT AND COST

Resilience

1. Meteorological-based Early Warning System: Typical cost range is **$0.5-7M** (according to World Meteorological Organization) for country-scale upgrades. Climate event losses in the country are around **1%** of their GDP annually (according to World Bank).
2. Corrosion resistant infrastructure: cost range is from **$20-600/m²** , but can save life cycle costs by **15-35%**
3. Pilot automated islanding software: cost range is **$100k-1M+** per pilot zone depending on size

Sustainability

Potential Plants that can be **Expanded** and **Why**

Pedro Corto Solar Plant

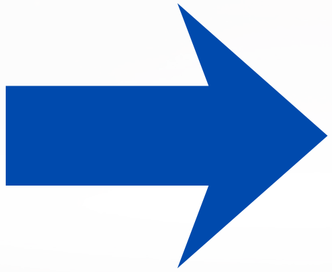
Calabaza 1 Solar Plant

Cotoperi Solar Complex

Pecasa Wind Farm

Biomass Energy Projects

El Guincho Photovoltaic Park



Excellent solar power radiation: **5 kWh/m²** per day

Consistent and strong wind conditions

High water resource potential

Can add **800-1000 MW** cumulatively

SUMMARY OF
OVERALL IMPACT

Resilience

- Hardened infrastructure cuts hurricane outages by **40–60%.**
- Early-warning improve response times by **30–50%** and save **$180M–$360M** per year
- Sources: CREWS Caribbean Early Warning Reports; WMO; SEL Inc.; Microgrid Knowledge.

Sustainability

- Solar and coastal wind expansion boosts renewables to **45–55% by 2040.**
- Hydro upgrades and biomass provide reliable, storm-resistant baseload power.
- Sources: NREL Renewable Potential Data; C2ES; Caribbean Biomass Studies.

Decentralisation

- Local energy systems reduce blackout duration by up to **90%.**
- Mapped renewable sites cut transmission losses by **8–12%.**
- **$96M–$144M** saved per year with a decentralised grid
- Sources: DOE Microgrid Research; Microgrid Knowledge; NREL Loss-Reduction Studies.

Economic & Social Impact

- Can create approx. **15,000–25,000** new jobs nationwide.
- Strengthens rural energy access and speeds recovery after storms.
- Sources: MDPI Microgrid ROI Study; DOE Community Energy Programs; C2ES.

A RESILIENT, SUSTAINABLE, DECENTRALISED ENERGY FUTURE FOR THE DOMINICAN REPUBLIC

Why this transformation matters

Frequent natural weather disasters, high distribution losses, and grid fragility make resilience, clean energy, and decentralization essential for the Dominican Republic to move forward as a nation and continue to develop.

What this plan delivers

- Reliable power during extreme storms, reducing widespread outages seen in hurricanes like Maria and Fiona
- Cheaper, cleaner electricity by scaling solar and coastal wind, lowering fuel-import dependence.
- Local energy autonomy through microgrids and renewable energy sites in high-resource regions like Montecristi.
- Long-term stability from fewer blackouts, lower O&M costs, and strong job creation.

A future-ready DR by 2040

By 2040, the Dominican Republic can have a resilient, sustainable, and decentralised energy system, providing reliable, low-carbon power that benefits communities, government, and energy companies alike.

THANK YOU